

RESEARCH INTERESTS

Tanya Lama is a National Science Foundation Postdoctoral Research Fellow in Biology focused on mammalian evolutionary and conservation genomics. Her research explores the genetic mechanisms underlying eco-evolutionary processes and factors influencing population persistence in natural and human-altered environments.

PROFESSIONAL APPOINTMENTS

Postdoctoral Research Fellow in Biology, National Science Foundation 2021 – present
Research Focus: rapid adaptation, life history trait evolution, lifespan~body size and lifespan~brain size allometries, genetic and epigenetic mechanisms underlying aging.
Co-PIs: Liliana Davalos (SUNY Stony Brook), Emma Teeling (U College Dublin)

NIH IRACDA Associate, National Institutes of Health 2021 – present

EDUCATION

Ph.D. Wildlife, *University of Massachusetts Amherst* 2021
M.S. Wildlife, *University of Massachusetts Amherst* 2017
B.S. Fisheries, *University of Connecticut* 2013

PEER REVIEWED PUBLICATIONS (* shared first-authorship)

- Forester B, **Lama TM*** (2022) “*The role of genomics in the future of ESA decision-making.*” *The Codex of the Endangered Species Act: The Next Fifty Years –Volume II.* Edited by Lowell E. Baird and John Organ, Rowman & Littlefield, July 2022, pp. 1-33. doi:10.32942/osf.io/b6rf4
- Cardoza JE, **Lama TM**, Mawdsley JR, Organ JF (2022) “*Canada lynx*” *Wild Furbearer Management and Conservation in North America.* Edited by Timothy Hiller, Publisher, January 2023, pp. 1-31.
- Lama, TM**, et al. (Accepted 2022) “Assessing genomic vulnerability among Canada lynx (*Lynx canadensis*) populations at the trailing edge.” *Molecular Ecology*.
- Dahn HA, et al. (2022) Benchmarking ultra-high molecular weight DNA preservation methods for long-read and long-range sequencing. *Gigascience*. 2022 Jun; 0:1-13. DOI: 10.1093/gigascience/giac068
- Moreno Santillan D, **Lama TM**, et al. (2021) Large-scale sampling reveals unique immunity and metabolic adaptations in bats. *Molecular Ecology*. 2021 Jun; 30:6449–6467.
- Rhie A, et al. (2021) Towards complete and error-free genome assemblies of all vertebrate species. *Nature*. 2021 Apr; 592 (7856):737-46.
- Organ JF, Decker TA, and **Lama TM** (2016) The North American model and captive cervid facilities: what is the threat? *Wildlife Society Bulletin* 40: 10-13 DOI: 10.1002/wsb.63
- Glinka CO, Frasca S, Provatas AA, **Lama TM**, DeGuise S and Bosker T (2015) The effects of model androgen 5 α -dihydrotestosterone on mummichog (*Fundulus heteroclitus*) reproduction under different salinities. *Aquatic Toxicology*, 165: 266-276.
- Organ JF, Decker DJ, Stevens SS, **Lama TM** and Doyle-Capitman C (2014) Public Trust Principles and Trust Administration Functions in the North American Model of Wildlife Conservation: Contributions of Human Dimensions Research, *Human Dimensions of Wildlife: An International Journal*, 19(5): 407-416

IN PREPARATION

1. **Lama TM**, Davalos LM, Borries C, Smaers JB. (2022) Lessons from small mammals: Lifespan allometries unveil the keys to exceptional longevity. *In Prep*.
2. Morales AE, **Lama TM**, Davalos LM, Hiller M (2022) Insights into immunity and viral tolerance are revealed through high-quality genomes of coronavirus reservoir bats. *In Prep*.

NON-REFEREED PUBLICATIONS

1. **Lama TM** (2019) New Genome Powers Canada Lynx Research & Conservation Efforts. *Massachusetts Wildlife*, 68(4): 22-25. *Print*.
2. Zarco D, Ilardo BS, Soriano L, **Lama TM**, Kleinman-Ruiz D, Godoy JA (2021) Report on the genetic monitoring of Iberian lynx reintroduction in Vale do Guadiana (2016-2020). Instituto da Conservação da Natureza e das Florestas.

EXTRAMURAL FUNDING

Sarah K. deCoizart Charitable Trust (\$20,000)	In review
Fulbright Scholar (Spain) (\$14,000)	2021
National Science Foundation Postdoctoral Research Fellowship in Biology (\$276,000)	2020
UMass Amherst Natural History Collection (\$3,996)	2018
Maine Department of Inland Fisheries and Wildlife (\$270,556)	2015

TEACHING & MENTORSHIP

NIH IRACDA Fellow, SUNY Stony Brook 2021 – present
An immersive teaching pedagogy and development program for underrepresented minority teacher-scholars in the biomedical sciences.

NSF Research Experience for Undergraduates, SUNY Stony Brook 2021
Co-directed an undergraduate research project on non-invasive genetic monitoring methods

Teaching Fellow, University of Massachusetts Amherst 2019
Led an undergraduate-level conservation biology course and completed pedagogy training.

SELECTED PRESENTATIONS & SEMINARS

1. **Lama TM** (2022) Evolutionary genomics for conservation: assessing vulnerability, resilience and response in a changing world. *Invited talk*. University of Michigan. Ann Arbor, MI.
2. **Lama TM** (2021) Exploring the bat immunome and the genomic underpinnings of extreme longevity. *Invited talk*. The Verena Consortium. Washington DC.
3. **Lama TM** (2021) Using gene-environment associations (GEA) to inform management in a changing environment. *Invited talk*. Conte Anadromous Fish Research Center. Amherst, MA.
4. **Lama TM** (2021) Assessing genomic vulnerability among Canada lynx (*Lynx canadensis*) populations at the trailing edge. *Invited talk*. California State University Stanislaus. Stanislaus, CA

5. **Lama TM** (2021) Using genomics to inform management of Canada lynx (*Lynx canadensis*) populations in a changing environment. *Invited talk*. Northern Arizona University. Flagstaff, AZ.
6. **Lama TM** (2019) Conservation Genomics & Wildlife Ecology, Dummerston Conservation Commission Winter Seminar Series. *Platform*. Dummerston, VT.

RESEARCH & RELATED WORK EXPERIENCE

Fulbright Scholar, Consejo Superior de Investigaciones Científicas, 2021

Co-developed non-invasive genetic monitoring strategies for the national Iberian lynx (*Lynx pardinus*) captive breeding and reintroduction program.

Research Assistant, Massachusetts Cooperative Fish & Wildlife Research Unit, 2017-2021

Released the first complete and annotated assembly of the Canada lynx (*Lynx canadensis*) genome. Assessed genome-wide diversity, connectivity, and adaptive potential across Maine and eastern Canada.

Contractor, US State Department, 2018

Prepared literature reviews on the environmental impacts of proposed infrastructure projects. Informed a policy framework which improves ecosystem management and attempts to mitigate illegal economic activities (illegal mining, illegal logging, wildlife trafficking) that support transnational criminal organizations.

Contractor, Department of Interior Northeast Climate Science Adaptation Center, 2017 – 2018

Investigated shifting distribution and abundance among red squirrel populations in the White Mountain National Forest in response to biennial mast and climate change.

Public Affairs Specialist, US Fish & Wildlife Service, 2015 – 2017

Engaged urban Hispanic audiences with the goal of increasing recreational use of the National Wildlife Refuge system. Co-supervised Hispanic Access Foundation interns at Priority Urban Wildlife Refuges.

Biological Science Technician, US Fish & Wildlife Service, Monomoy National Wildlife Refuge, 2014 Identified, trapped, banded, and monitored productivity of federally listed shorebirds. Assisted in the capture, handling, and tagging of shorebirds and songbirds using cannon and mist nets.

Biological Science Technician, Wildlife & Sport Fish Restoration, US Fish & Wildlife Service, 2013 Participated in grant approval and accomplishment reporting for restoration and monitoring projects.

Research Assistant, Elephants Without Borders & San Diego Zoo Global, 2013 – 2015

Evaluated stress-hormone expression among African elephants working in the elephant-back tourism industry. Translated research findings into a policy recommendation to outlaw elephant-back tourism in Botswana in 2016.

Research Assistant, UConn Center for Environmental Science and Engineering, 2011 – 2013

Conducted aquatic toxicology research on endocrine disrupting (androgen mimicking) substances. Used enzyme-linked immunosorbent assays to quantify testosterone and estradiol hormone concentrations in fish during a standard 21-day exposure to a model androgen

PROFESSIONAL AFFILIATIONS

Northeast Climate Science Adaptation Center, Genome 10K Community of Scientists, Vertebrate Genomes Project, Bat1K, The Wildlife Society

ACADEMIC SERVICE

Ad Hoc Reviewer: *Conservation Physiology*, *Peer J Life & Environment*, *Evolutionary Anthropology*, *Molecular Ecology*, *J Experimental Biology*, *Evolutionary Applications*

REFERENCES

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Chief Emeritus, Cooperative Fish & Wildlife Research Units
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